前提：集总电路

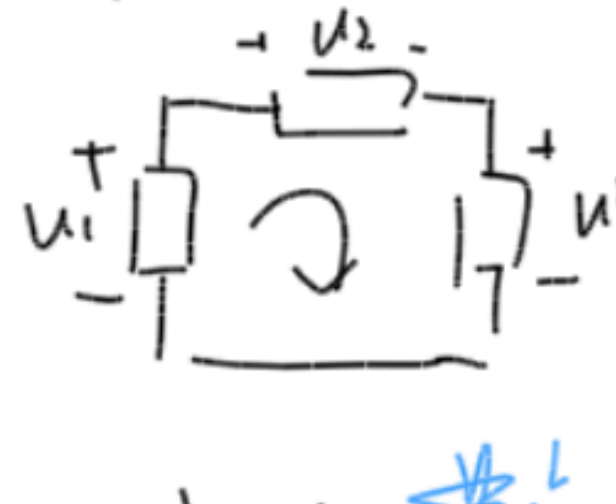
1. 参考方向：

关联：$P=ui>0$ → 消耗P

非关联：$P=-ui>0$ → 提供P

2. 基尔霍夫定律（KVL+KCL）→ 统一参考方向!!!

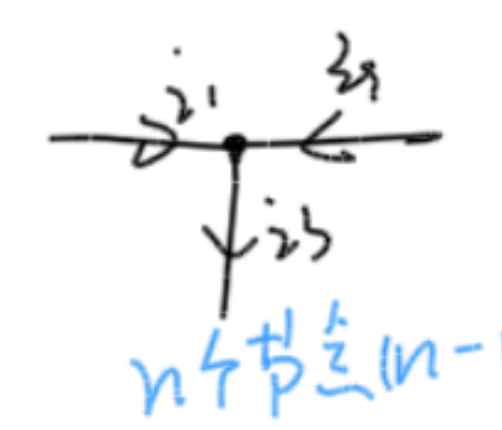
① KVL：回路



$u_1+u_2+u_3=0$

$\Sigma u=0$

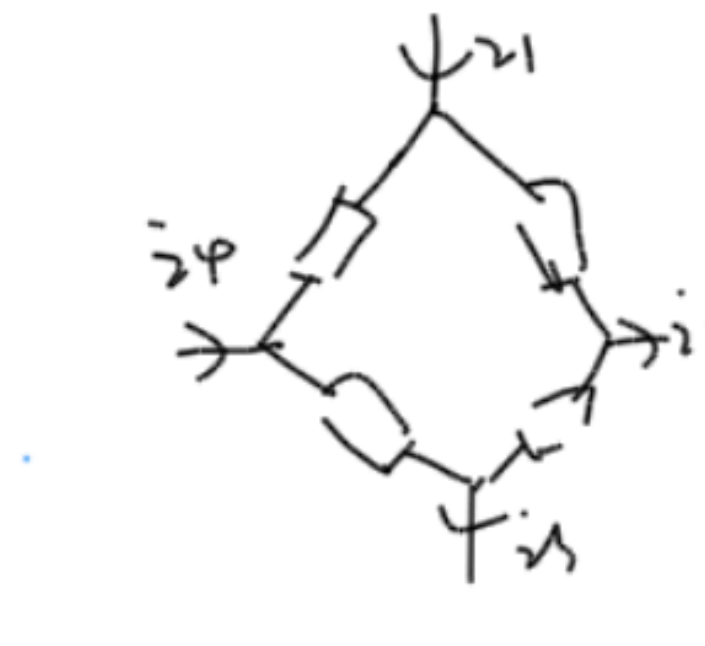
② KCL 节点　　对闭合面也成立



$i_1+i_2=i_3$

$\Sigma i_{in}=\Sigma i_{out}$

n个节点(n-1)个KCL方程

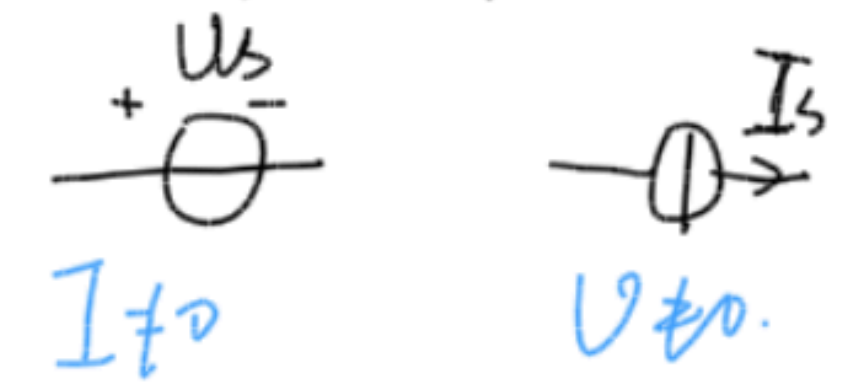


3. 电阻：

电导：$G=\frac{1}{R}$（西门子，S）

$i=Gu$

4. 电压源/电流源



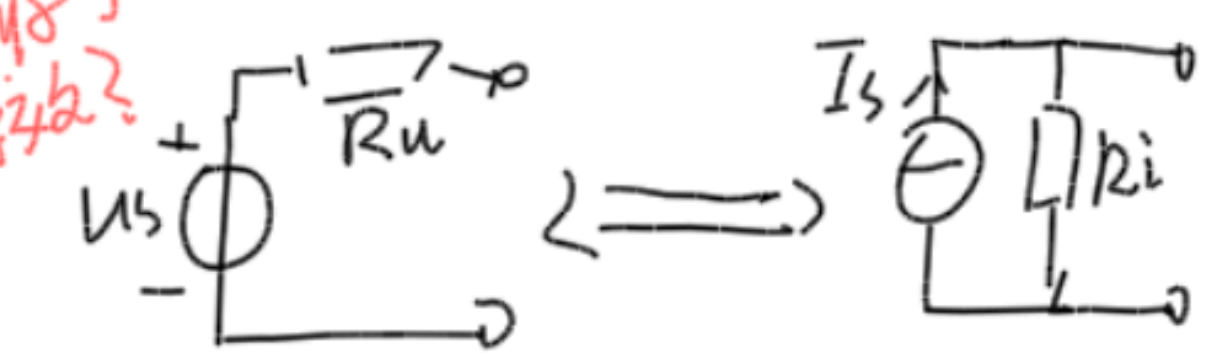
I任意　　U任意

不同恒压源不能并联

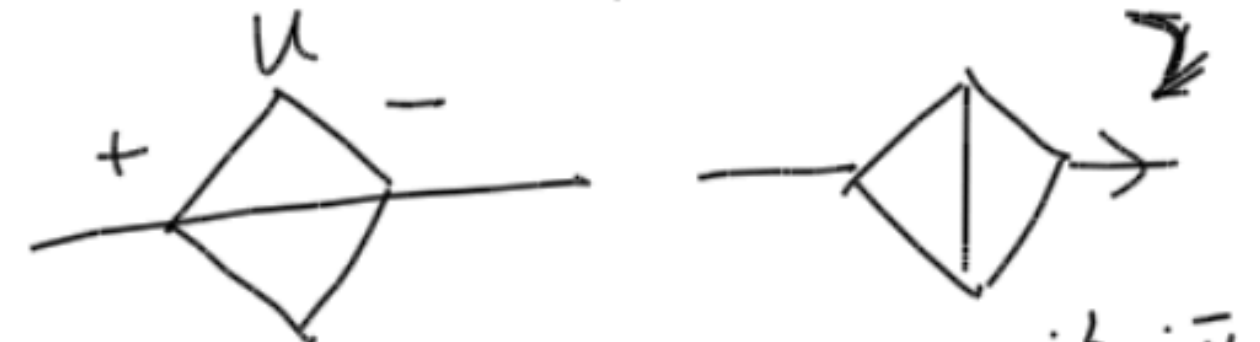
不同恒流源不能串联

等效：当 $R_u=R_i$，$U_S=I_S\cdot R_i$

受控源可以等效？



5. 受控源



～电压源　　～电流源

μu_1：电压控　μi_1：电流控

6. 分压与分流公式

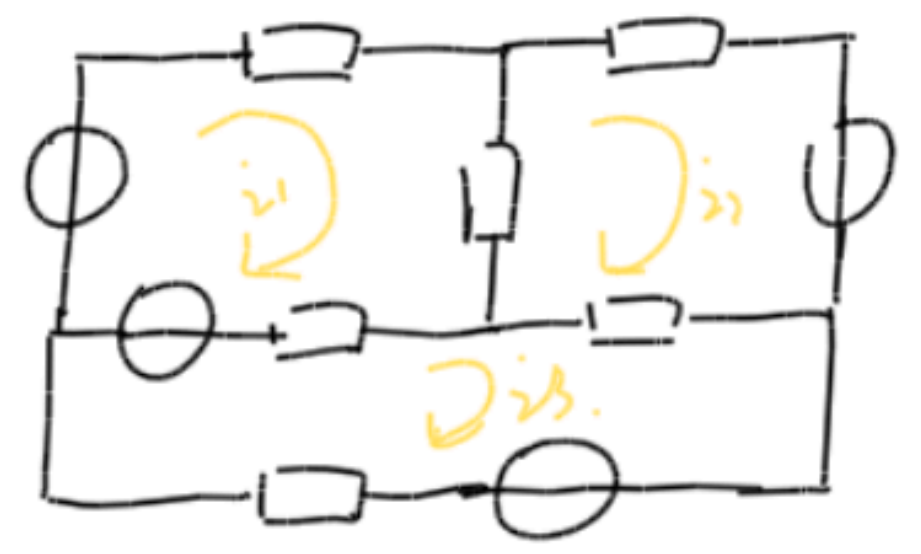
$u_k=\frac{R_k}{\sum_{i=1}^{n}R_i}u$，　$i_k=\frac{G_k}{\sum_{i=1}^{n}G_i}i$

7. 网孔分析：设网孔电流

自电阻 × 网孔i + 互电阻 × 相邻网孔i

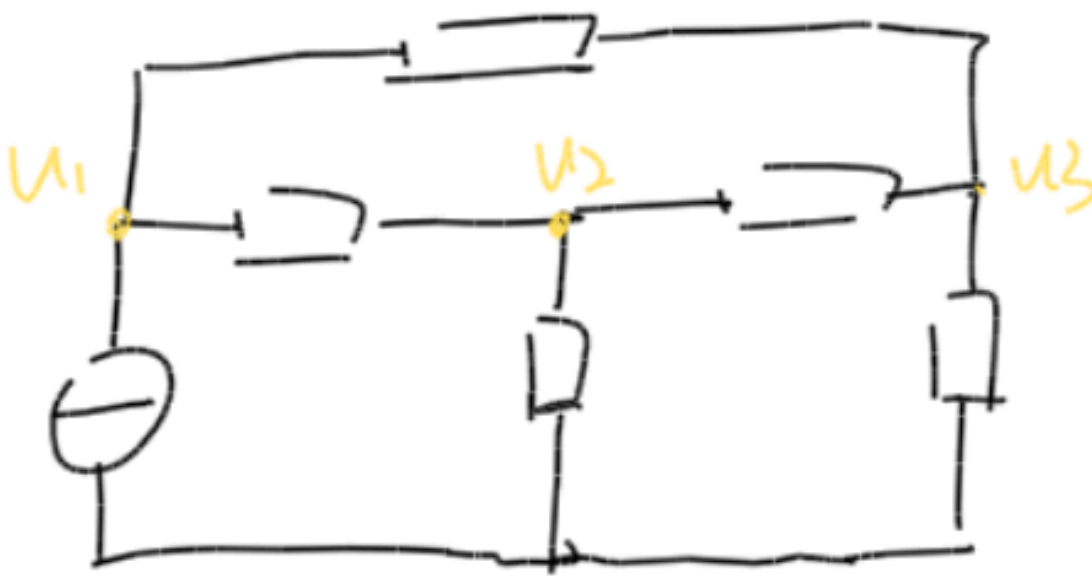
= 电压源升电压 → 注意参考方向

若电流源在边路：$i=i_S$



8. 节点分析 → 可由KCL推导得

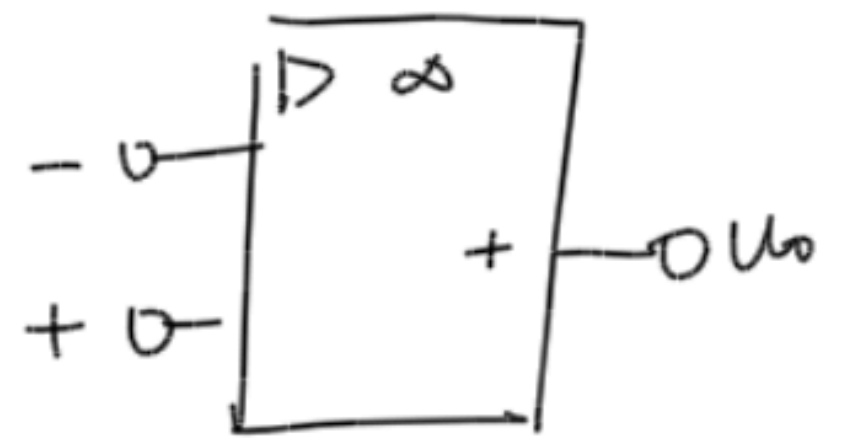
自电导 × 节点u − 互电导 × 相邻节点u = 流入电流源的代数和



可以通过接地减少方程数

n个节点，b个支路，b−n+1个独立方程

9. 运放电路



理想运放：$u_-\approx u_+$，$i\approx 0$

虚短虚断

⇓

原因：$A\to\infty$

做题 ☆

10. 三极管

外大内小

$R_\Delta=3R_Y$

11. 比例性

单一激励的线性、时不变电路

$H=\frac{响应}{激励}$

12. 叠加原理

有多个源的时候，置零

u→短；i→断

后叠加　　注意方向

功率不能叠加

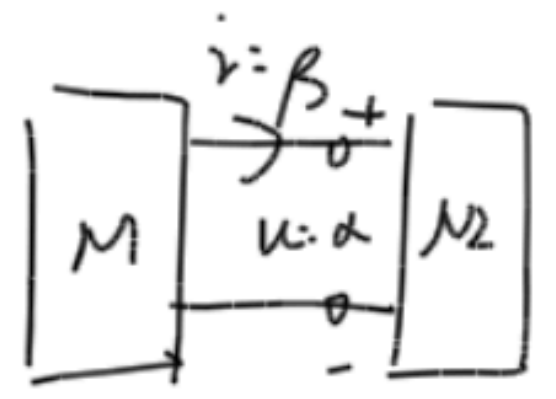
13. 分解基本方法

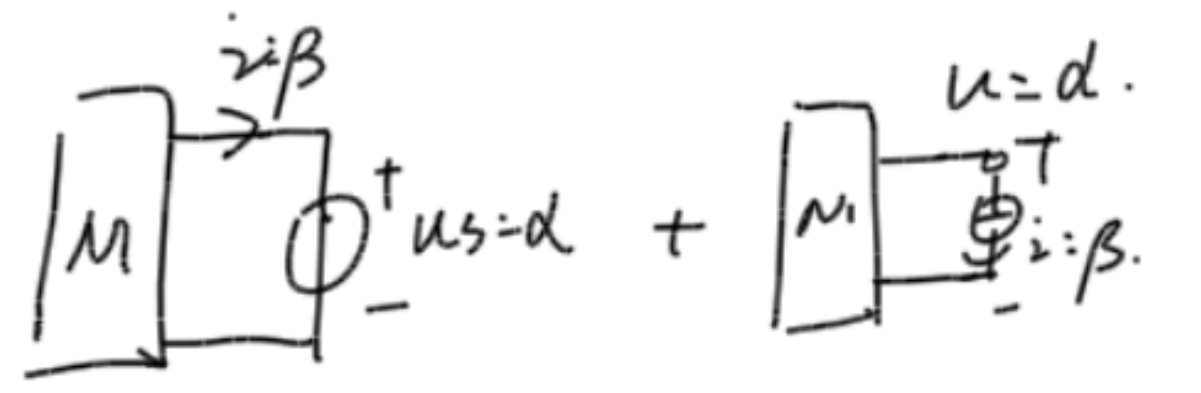
① 分为N1, N2，确定交处u, i方向

② 求VCR

③ 求内部u, i

14. 置换定理



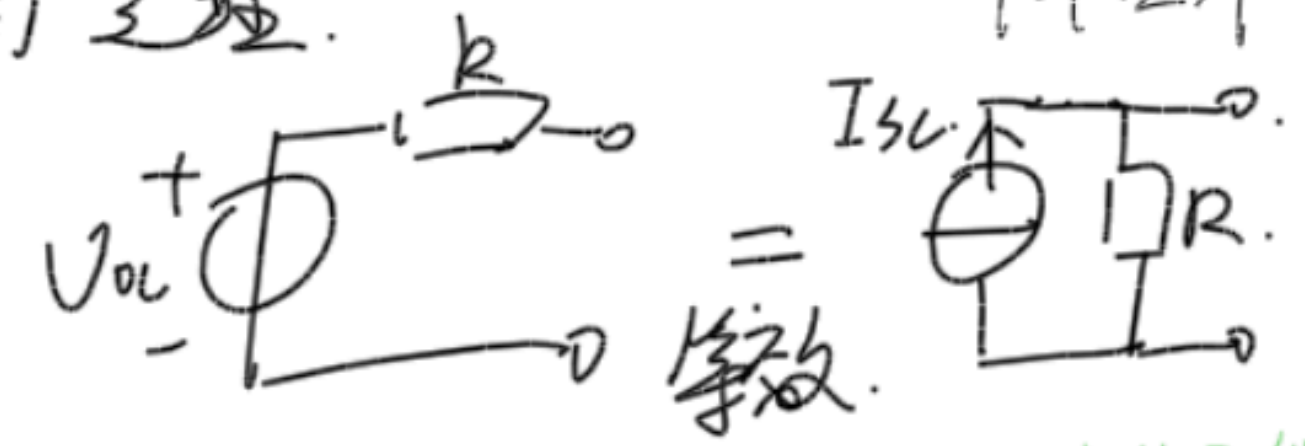


15. 戴维南/诺顿定理

U_{OC}：开路电压

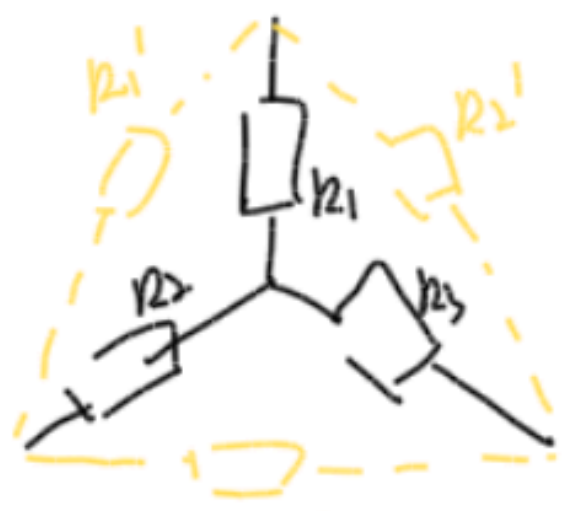
I_{SC}：短路电流

R：等效内阻　$U=U_{OC}-IR$



与⊖并，与⊕串的元件可去。两个相同的电压源并可以等效为一个

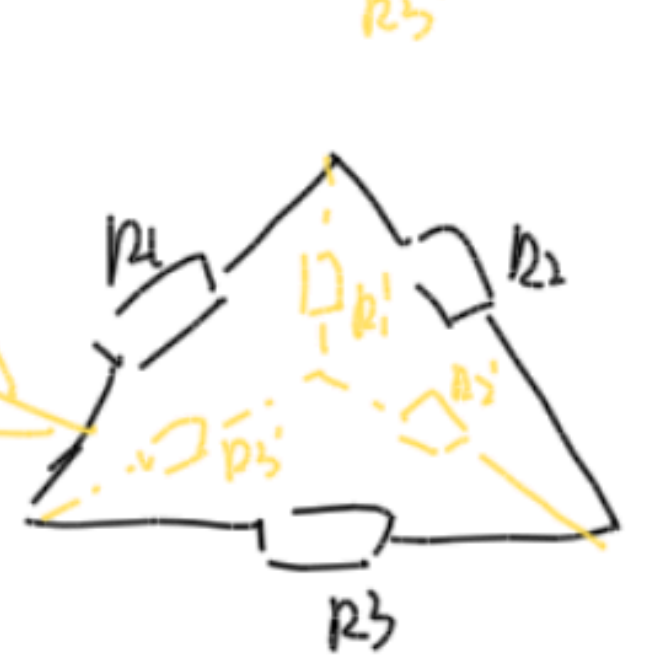
16. T形与Π形的变换



$R_1'=\frac{R_1R_2+R_2R_3+R_3R_1}{R_3}$

$R_2'=\frac{R_1R_2+R_2R_3+R_3R_1}{R_2}$

$R_3'=\frac{R_1R_2+R_2R_3+R_3R_1}{R_1}$



$R_1'=\frac{R_1R_2}{R_1+R_2+R_3}$

$R_2'=\frac{R_2R_3}{R_1+R_2+R_3}$

$R_3'=\frac{R_1R_3}{R_1+R_2+R_3}$

17. 电容　$\frac{1}{C_串}=\sum\frac{1}{C}$　$C_并=\sum C$

$q(t)=Cu(t)$，

$i(t)=C\frac{du(t)}{dt}$

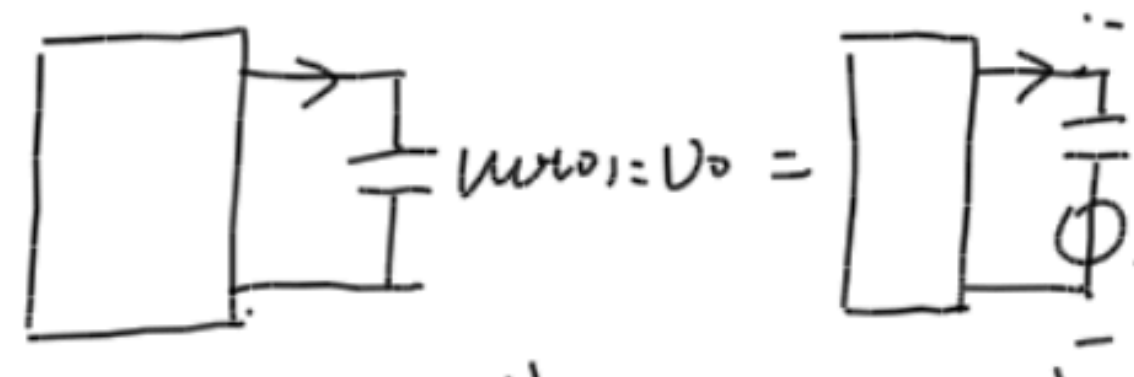
$e(t)=\frac{1}{2}CU^2$

隔直流

u(t)不突变：$u(t_-)=u(t_+)$

$u(t)=u(t_0)+\frac{1}{C}\int_{t_0}^{t}i(\xi)d\xi$

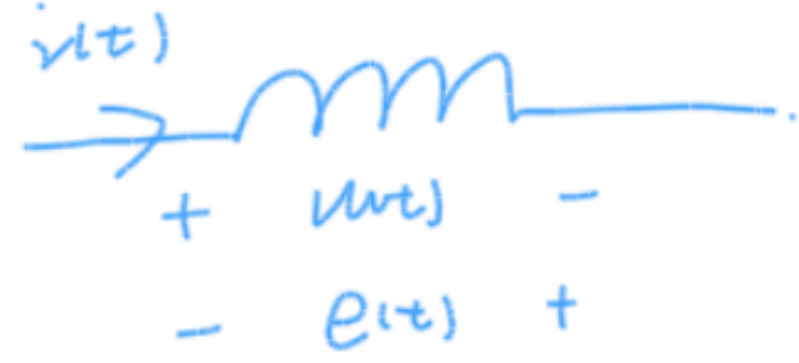
记忆性质　　等效



18. 电感

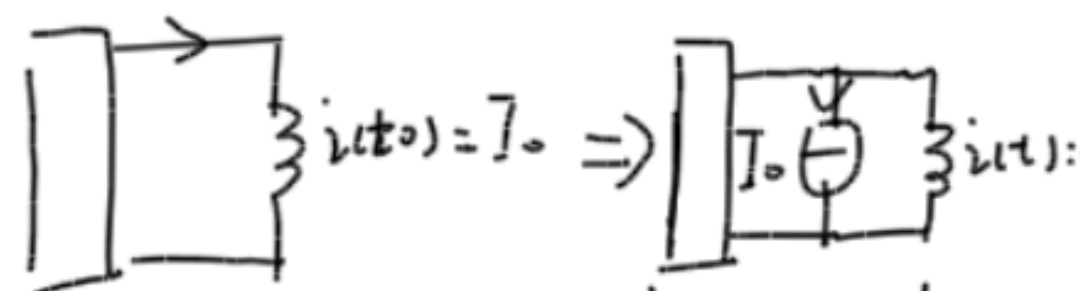
$u(t)=L\frac{di(t)}{dt}=-e$

$i_L(t)=i_L(t_0)+\frac{1}{L}\int_{t_0}^{t}u(\xi)d\xi$



$e(t)=\frac{1}{2}Li^2(t)$

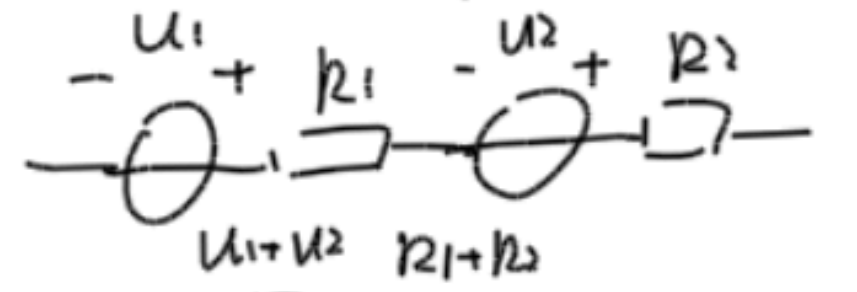
i(t)不突变：$i(t_-)=i(t_+)$

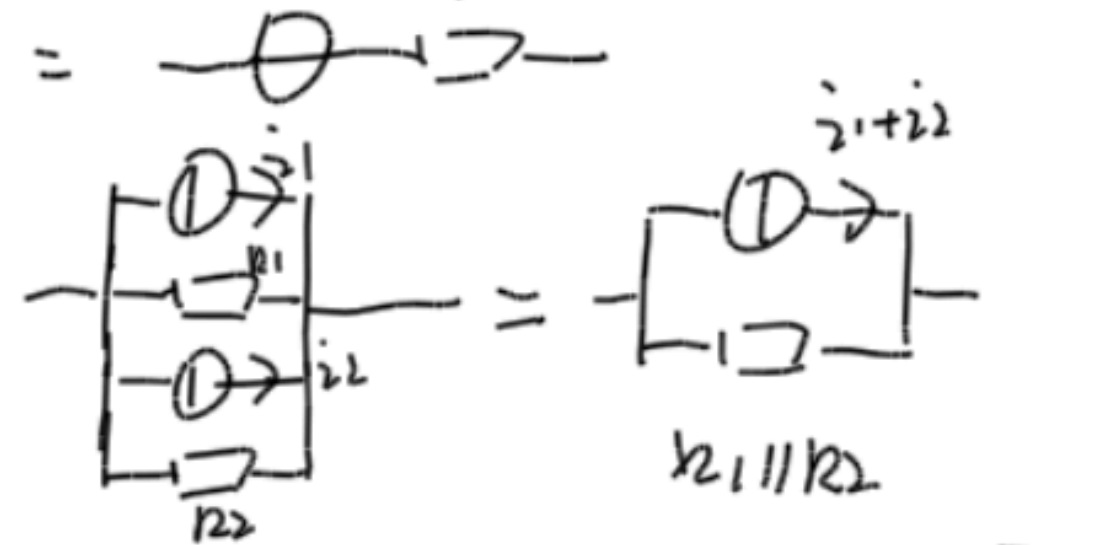


$L_串=\Sigma L$，$\frac{1}{L_并}=\Sigma\frac{1}{L}$

零状态响应 { 单位阶跃响应，单位阶跃in ε(t)；单位冲击响应，单位冲击in δ(t) }

19. 电压源/电流源等效

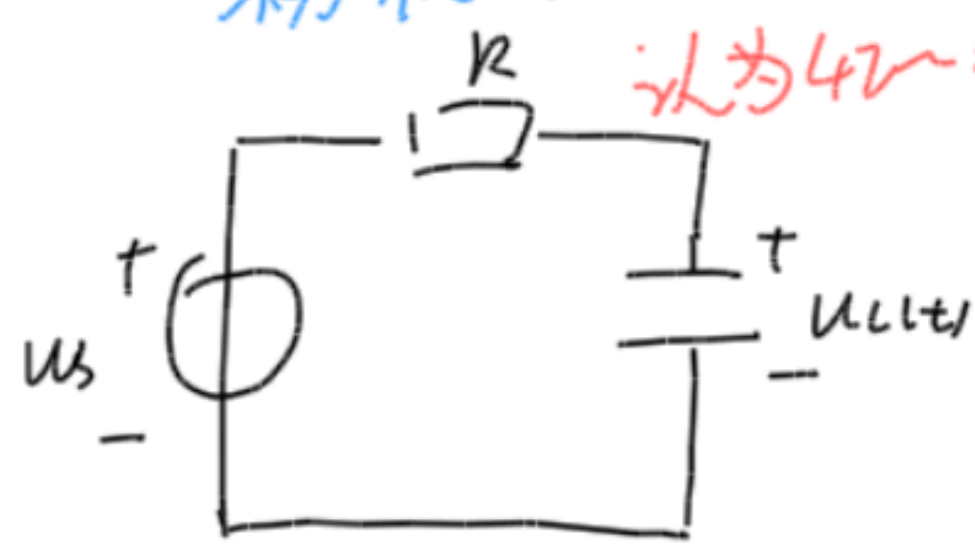




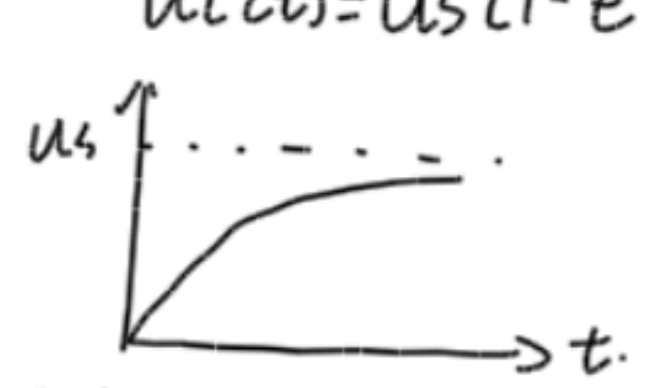
19. 零状态响应

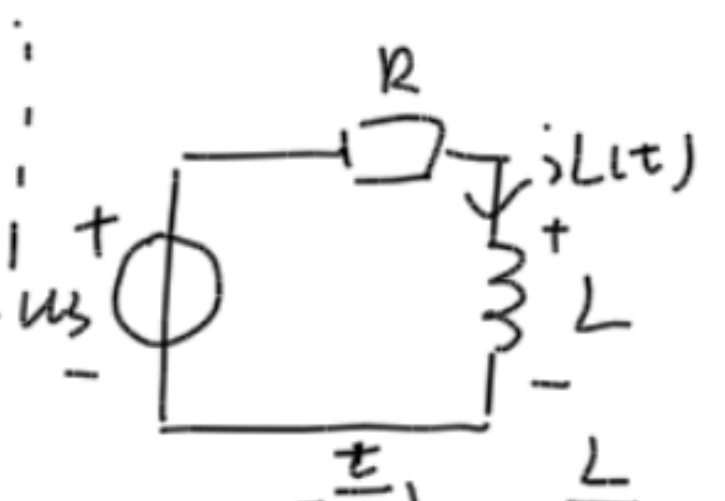
初状态为0，$u_C(t_0)=0$

认为4τ~5τ稳定

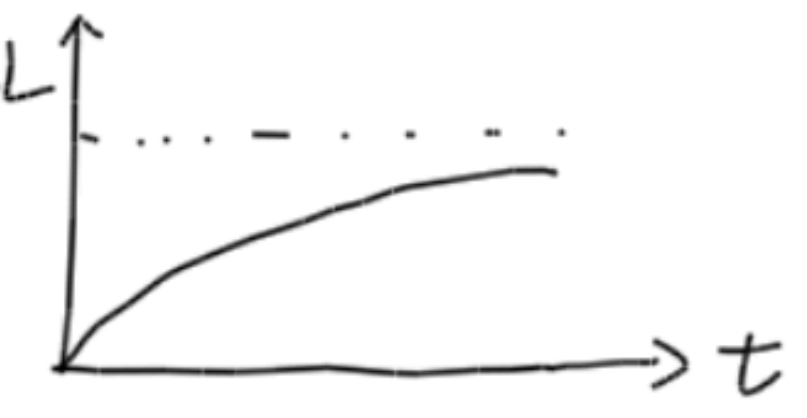


$u_C(t)=U_S(\infty)(1-e^{-\frac{t}{\tau}})$，$\tau=RC$





$i_L(t)=i_L(\infty)(1-e^{-\frac{t}{\tau}})$，$\tau=\frac{L}{R}$



20. 阶跃响应

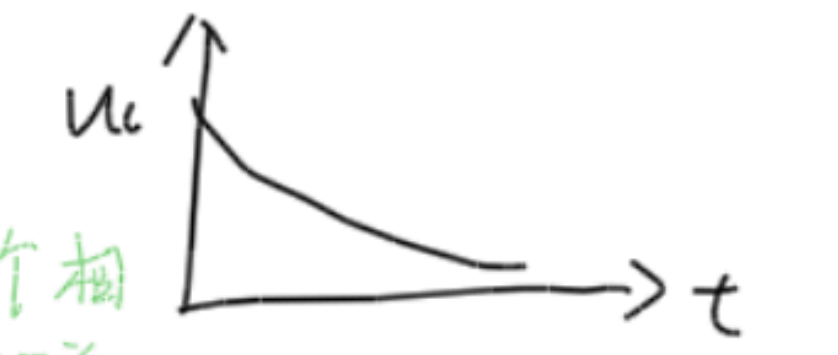
$\varepsilon(t)=\begin{cases}0, & t<0\\1, & t>0\end{cases}$；$\varepsilon(t-t_0)\begin{cases}0, & t<t_0\\1, & t>t_0\end{cases}$

单位冲击函数：$\delta(t)=\frac{d\varepsilon(t)}{dt}=\begin{cases}0, & t\neq0\\\infty, & t=0\end{cases}$

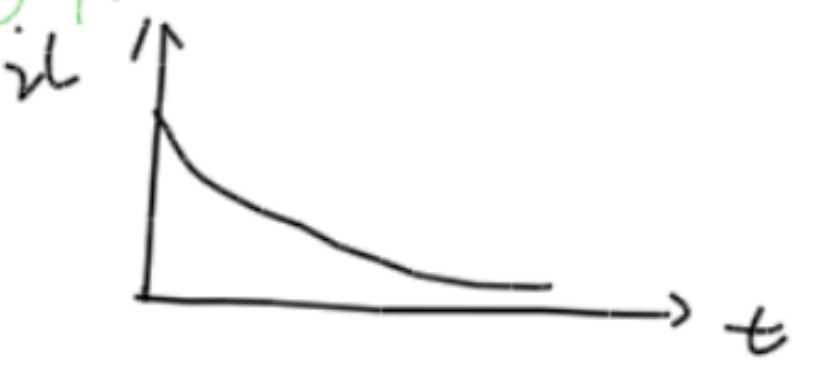
$f(t)\delta(t)=f(0)\delta(t)$ → 任何函数与δ(t)相乘都等于该函数在t=0处乘以δ(t)

21. 零输入响应 → 放电

$u_C(t)=u_C(0)e^{-\frac{t}{\tau}}$



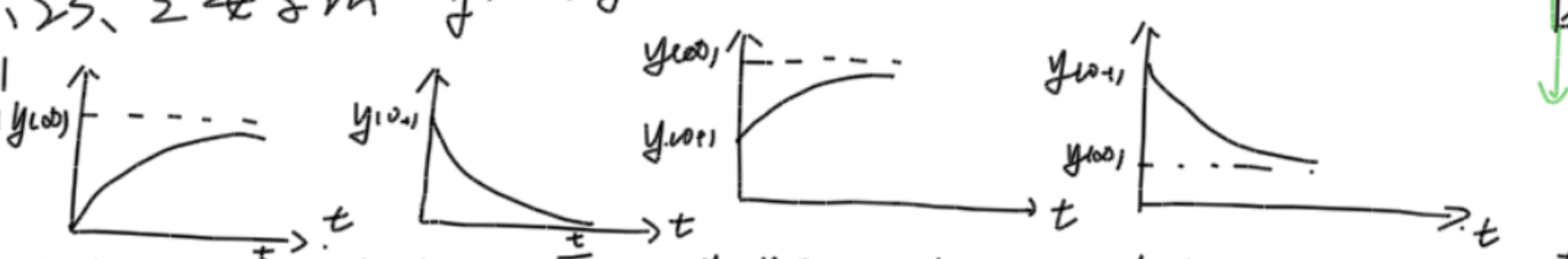
$i_L(t)=i_L(0)e^{-\frac{t}{\tau}}$



22. 叠加原理

全响应 = 零状态 + 零输入

23. 三要素法　$y(0_+)$，$y(\infty)$，τ



$y=y(\infty)(1-e^{-\frac{t}{\tau}})$　$y=y(0_+)e^{-\frac{t}{\tau}}$　$y=y(0_+)+(y(\infty)-y(0_+))(1-e^{-\frac{t}{\tau}})$　$y=y(\infty)+[y(0_+)-y(\infty)]e^{-\frac{t}{\tau}}$

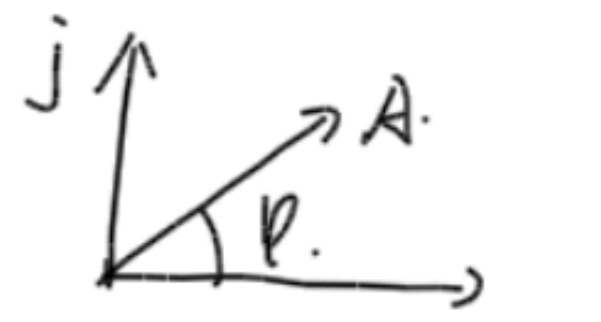
24. 瞬态与稳态

响应 $y(t)=y(\infty)+[y(0_+)-y(\infty)]e^{-\frac{t}{\tau}}$

稳态响应　　瞬态响应

25. 相量

$\dot{A}=A\angle\varphi=A\cos\varphi+jA\sin\varphi$

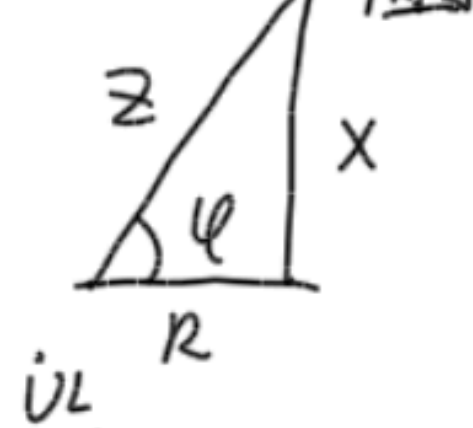


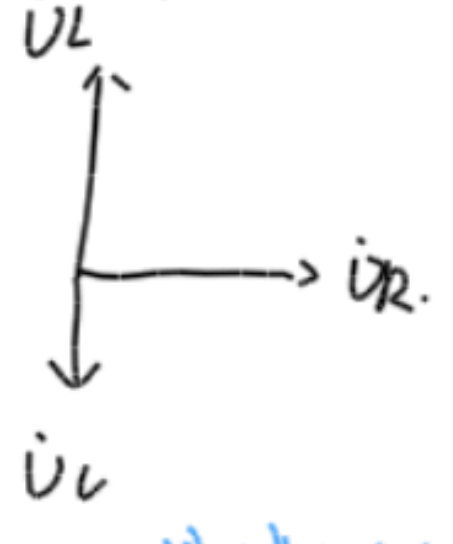
26. 阻抗与导纳

	Z	$Y=\frac{1}{Z}$
R	R	
C	$\frac{1}{j\omega C}$	
L	$j\omega L$	

$\dot{U}_m=Z\dot{I}_m$

阻抗三角形





在相量模型中前面的分析方法都成立

27. 有效值

$I=\sqrt{\frac{1}{T}\int_0^T i^2dt}$

U同理

28. 相量图法

可以利用相量图解题

29. 功率　增大cosφ，并联电容

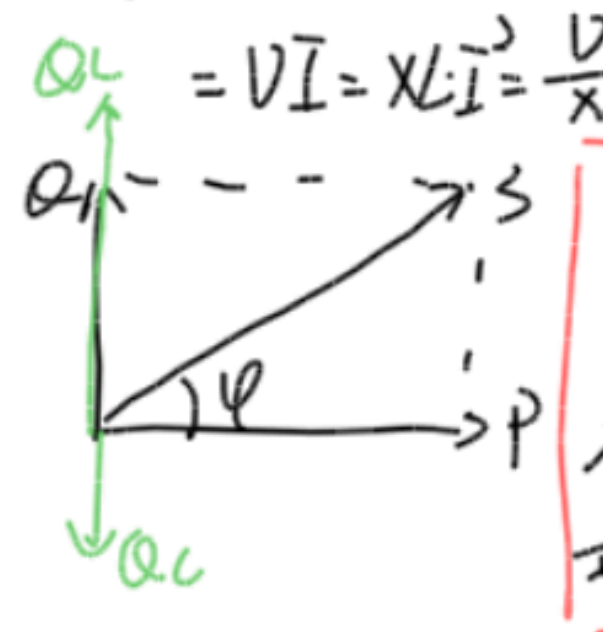
平均功率：$P=U_有\cdot I_有$　有功功率

无功功率 → 电源与电容/电感之间往返的P

= 瞬时功率的最大值

$Q_L=2\omega W_L$ ⊕，$Q_C=-2\omega W_C$ ⊖

$=UI=X_LI^2=\frac{U^2}{X_L}$　$=-UI=-X_CI^2=-\frac{U^2}{X_C}$



有功功率 $P=UI\cos\varphi$
无功功率 $Q=UI\sin\varphi$
视在功率 $S=UI$
功率因数 $\lambda=\frac{P}{S}=\cos\varphi$

$\varphi=\varphi_u-\varphi_i$

30. 复功率

$\bar{S}=\dot{U}\cdot\dot{I}^*$

$=P+j2\omega(W_L-W_C)$

31. 最大功率传递定理

恒源：$P_{max}=\frac{U_{OC}^2}{4R_0}=\frac{I_{SC}^2}{4G_0}$

正弦稳态：

① $\dot{U}_{OC}$、Z_0不变，共轭匹配

$Z_S=Z_L^*$，$P_{max}=\frac{U_S^2}{4R_S}$

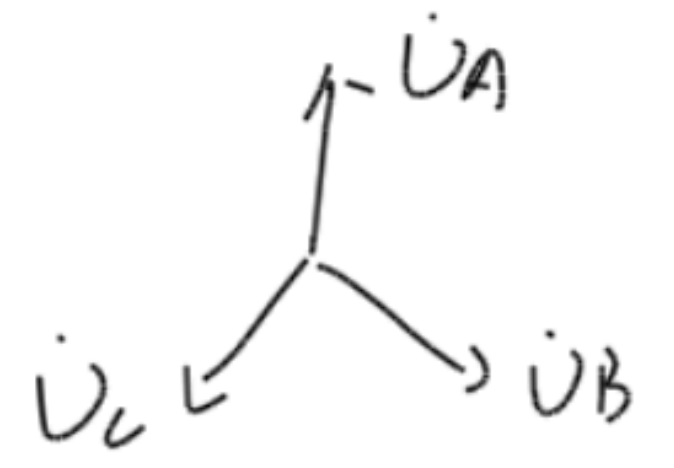
② 纯电阻，模匹配

$R_L=|Z_0|$，$P_{max}=I^2R_L$

③ φ固定，模匹配

$|Z_L|=|Z_0|$

31. 三相电路



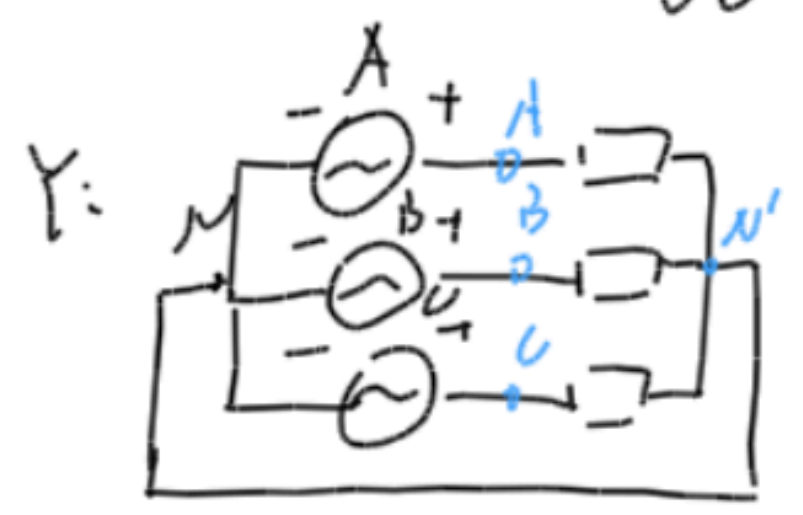
$|\dot{U}_A|=|\dot{U}_B|=|\dot{U}_C|$

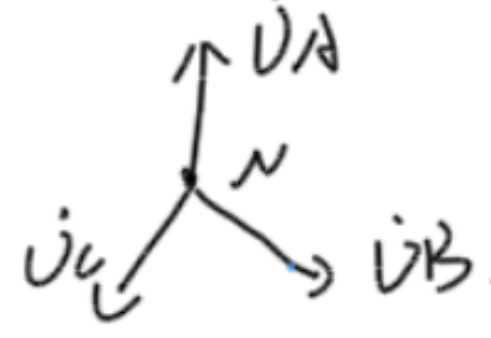
$\dot{U}_A=U\angle0°$

$\dot{U}_B=U\angle-120°$

$\dot{U}_C=U\angle120°$

Y：





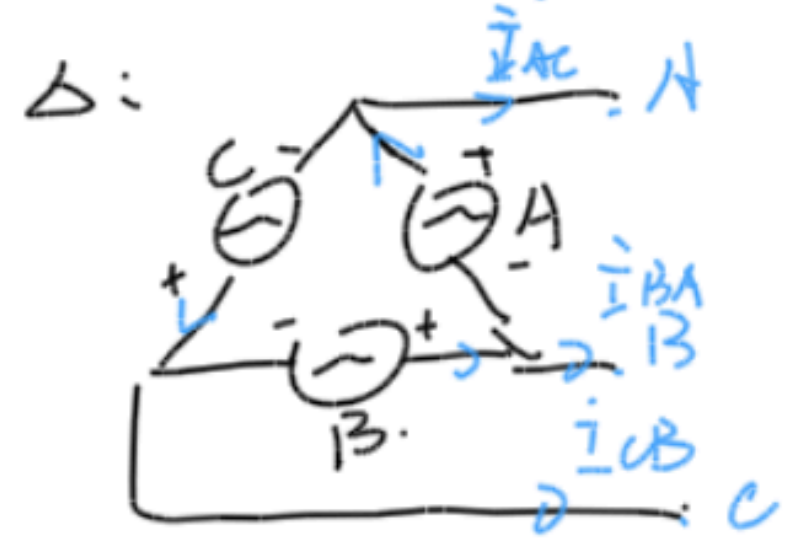
$\dot{U}_{AB}=\dot{U}_A-\dot{U}_B$　$I_{NN'}=0$

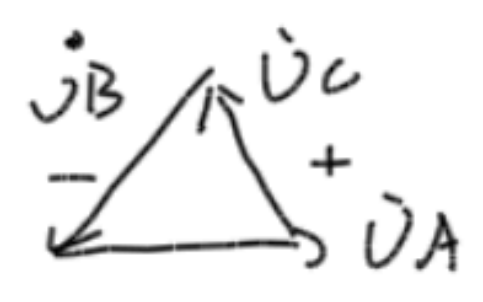
$\dot{U}_{BC}=\dot{U}_B-\dot{U}_C$

$\dot{U}_{CA}=\dot{U}_C-\dot{U}_A$

→ KVL可解

Δ：





$\dot{U}_{AB}=\dot{U}_A$，$\dot{U}_{BC}=\dot{U}_B$，$\dot{U}_{CA}=\dot{U}_C$

$\dot{I}_{AC}=\dot{I}_A-\dot{I}_C$，$\dot{I}_{BA}=\dot{I}_B-\dot{I}_A$

$\dot{I}_{CB}=\dot{I}_C-\dot{I}_B$

$P=3U_pI_p\cos\varphi$ → 相电压与相电流的相位差角

不对称：$P=P_A+P_B+P_C$；$P_A=U_{Ap}I_{Ap}\cos\varphi_A$

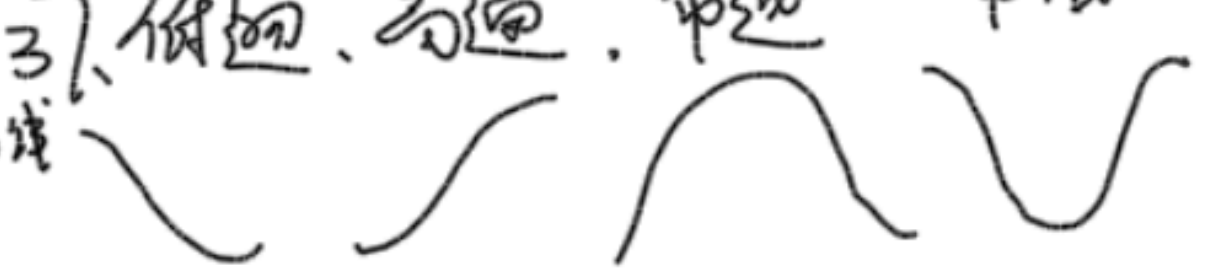
35. 截止频率

当 $|H_u(\omega)|=\frac{1}{\sqrt{2}}|H_{um}|$，

$\omega_C=\frac{1}{\tau}$ 截止频率

0~ω_C，通频带

37. 低通、高通、带通　带阻



32. 多频正弦稳态电路

$Z=|Z|\angle\varphi$

$\dot{U}$、$\dot{I}$的比值　相位关系

$Z(j\omega)=R(\omega)+jX(\omega)$

$=|Z(j\omega)|\angle\varphi(\omega)$

幅频特性　相频特性

频率特性

输入阻抗 $Z(j\omega)$ 可以看作 $1\angle0°$A激励时所产生的电压响应

非正弦稳压 $\xrightarrow{傅里叶变换}$ 多个正弦+直流

⇓

叠加原理

有效值 $I=\sqrt{\Sigma I^2}$，$U=\sqrt{\Sigma U^2}$

频率叠加：多个频率不同且比为有理数的正弦电压/电流产生的平均功率等于单独产生的平均功率之和。同频基波可叠加。

33. RLC电路的谐振

$\dot{U}$与$\dot{I}$同相位　注意单位 ω → rad/s，f → s^{-1}/Hz　串 ⇒ 短　并 ⇒ 开

条件：$X_L=X_C$

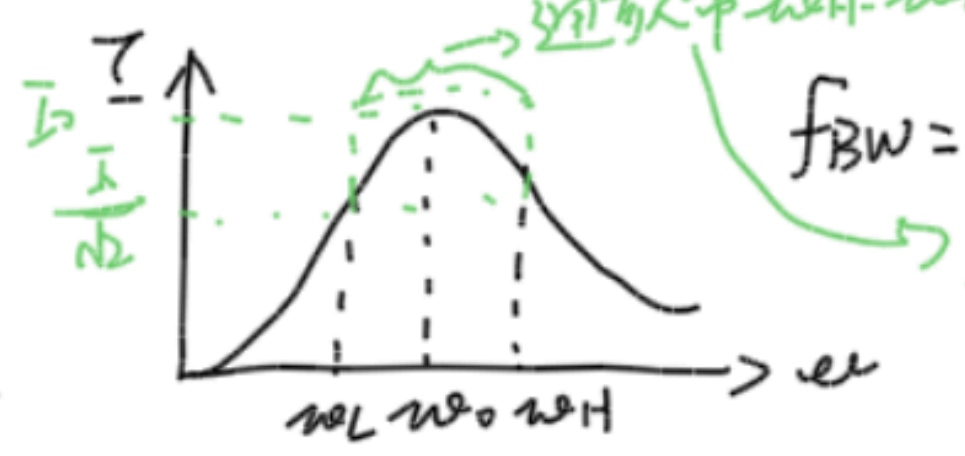
$2\pi fL=\frac{1}{2\pi fC}$ ⇒ $f=\frac{1}{2\pi\sqrt{LC}}$，$\omega_0=\frac{1}{\sqrt{LC}}$

品质因数：$Q=\frac{1}{R}\sqrt{\frac{L}{C}}=\frac{\omega_0L}{R}$　$Q=\frac{U_L}{U}=\frac{U_C}{U}$

$U_C=U_L=QU_R=QU$

并联同理

通频带



$f_{BW}=\frac{R}{2\pi L}=\frac{f_0}{Q}$

Q↑，f_{BW}↓，选择性越好

34. 阻尼电阻 $s_{1,2}=-\frac{R}{2L}\pm\sqrt{(\frac{R}{2L})^2-\frac{1}{LC}}$

串 $R_d=2\sqrt{\frac{L}{C}}$　并 $G_d=2\sqrt{\frac{C}{L}}$

$R>R_d$，过阻尼，$s_{1,2}$为不相等的负实数 $=-\alpha_1,-\alpha_2$，$u_C(t)=k_1e^{-\alpha_1t}+k_2e^{-\alpha_2t}$

$R=R_d$，临界，$s_{1,2}$为相等的负实数 $=-\alpha$，$u_C(t)=k_1e^{-\alpha t}+k_2te^{-\alpha t}$

$R<R_d$，欠阻尼，$s_{1,2}$为共轭复数，实部为负数 $s_1,s_2=-\alpha\pm j\omega_d$，$u_C(t)=Ke^{-\alpha t}\cos(\omega_dt+\varphi)$

$R=0$，无阻尼，$s_1,s_2=\pm j\omega_d$，$u_C(t)=K\cos(\omega_dt+\varphi)$